



# Environmental impacts caused by food waste: A case study in Brazilian supermarkets

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## ARTICLE INFO

### Keywords:

Food waste  
Fruits and vegetables  
Supermarket  
Waste composition analysis  
LCA  
Carbon footprint

## ABSTRACT

The growing global demand for food, alongside with an increase in the environmental pressures and food insecurity, highlights the urgency of addressing food loss and food waste. This study investigates the magnitude and environmental impact of fruit and vegetable waste in three Brazilian supermarkets located in the State of São Paulo. Using waste composition analysis and life cycle assessment, 2160.3 kg of unsold fruit and vegetables were quantified over one week per store, corresponding to an estimated 2366.3 kg of carbon dioxide-equivalent emissions. The products with the highest waste were bananas, tomatoes, cabbage, potatoes, and oranges — accounting for nearly half the total waste mass. Emissions were highest for fruits, followed by vegetables, and tubers, bulbs and roots. Unlike many studies that rely on secondary data or literature, this research provides original, primary data from supermarket waste, offering valuable insights into retail-level food waste and its environmental consequences. The study emphasizes the importance of targeting high-impact waste categories and implementing context-specific interventions, including improved cold chain infrastructure, employee training, and flexible product specifications. Findings offer actionable insights for retailers to reduce food loss and food waste and associated emissions, contributing to national and international sustainability goals. The integration of waste quantification with life cycle assessment represents a robust approach to identifying environmental hotspots and informing food loss and food waste mitigation strategies in supermarket retail.

## 1 Introduction

The global population is expected to reach 9 billion in 2050 which will compel countries to expand 60 % of their food production capacity and lead to a 40 % increase in the use of water and depletion of other natural resources on the planet (FAO, 2015). An increase in world hunger and the environmental impacts affecting the planet has led organizations to adopt innovative sustainable actions by aligning their strategies in accordance with the challenging demands of the consumer market and achieving greater competitiveness, given an increasingly competition thereof.

The last decade, the planet's temperature has already experienced a 1.1 ° Celsius increase and will have probably exceeded 1.5 ° by 2030. Such an increase in temperature, mainly caused by the emission of greenhouse gases (GHG), will significantly affect food production and, consequently, the availability of food worldwide (UNEP, 2022).

In this context, one of the actions that can contribute to reducing this impact is the prevention and reduction of food waste, which, in addition

to reducing carbon emissions through the disposal of unused food, makes the production, distribution and consumption of food more efficient through the rational use of resources.

These impacts can be minimized with actions that prevent the generation of FLW (Food Loss and Waste). Measuring the direct quantities of wasted food assists in identifying the carbon footprint from their production to disposal. In addition, the Life Cycle Assessment (LCA) is a tool widely used to calculate the environmental footprints of the entire life cycle of food (Kucukvar et al., 2019), or at least part of it, well capable of aiding companies to focus more closely on the fact that such products cause greater environmental damage.

Based on this context, this study aims to quantify the food wasted in three Brazilian supermarkets in the state of São Paulo, in terms of mass, and estimate the climate impact generated by the total waste identified in the stores, through the development of an LCA. Food waste prevention goals - such as enhancing food security, reducing costs, or minimizing environmental impacts - require tailored measurement. Therefore, to reduce environmental impacts, assessments should

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<https://doi.org/10.1016/j.clwas.2026.100468>

Received 22 May 2025; Received in revised form 2 October 2025; Accepted 2 January 2026

Available online 3 January 2026

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prioritize high-impact waste fractions, targeting those where reductions would yield the greatest savings. Identifying "low-hanging fruit" for analysis and prevention is crucial for effective interventions.

In order to better understand the difference between food loss and food waste, the FAO definitions were adopted in this work: "food loss is the decrease in the quantity or quality of food resulting from decisions and actions of food suppliers in the chain, excluding retailers, food service providers and consumers", while food waste is the "decrease in the quantity or quality of food resulting from decisions and actions of retailers, food service providers and consumers" (FAO, 2019, p. 10).

This research has as its primary aim to describe food waste not only in terms of mass but also in relation to its environmental impacts to support the prevention and management of food waste in Brazilian supermarkets, a critical area that has not been extensively explored through waste quantification and LCA. Waste prevention consists of measures that are adopted to prevent or reduce the causes of the problem, which is waste, to avoid the generation of waste and reduce food surplus as a primary conduct (NRDC, 2020). This study integrates waste quantification with LCA, particularly focusing on carbon footprint indicators, to identify hotspots to be prioritized in the investigation of waste generation causes and the development of prevention and valorization measures.

## 2 Literature review

### 2.1 Production and waste of FV in Brazil and the carbon footprint

In 2022, the total amount of fruit and vegetables sold in the main Supply Centers in Brazil was 17.5 million tons, representing R\$61.8 billion for the Brazilian economy (CONAB, 2023).

The State of Sao Paulo stands out as the largest producer, in addition to offering the greatest diversity of vegetables in the country. The 2017 harvest corresponded to 4.19 million tons of 53 species of these products in a cultivation area of 156.77 thousand hectares in the State. Orange is the most typically produced fruit in Brazil, followed by bananas, potatoes and tomatoes (IBGE, 2022). Tomatoes are the most frequently consumed vegetables in the country, followed by potatoes, onions and carrots. Bananas, oranges, watermelons, apples and papayas are the most typically consumed fruits (CES/IBGE (2017–2018).

Brazil is a major producer and consumer of vegetables. However, this also results in a proportionally significant amount of waste. Brazil is among the countries with the highest levels of food waste globally. It is estimated that 37 million tons of food are discarded in Brazil each year (Embrapa, 2018). It is also estimated that post-harvest losses of fruits and vegetables range between 35 % and 55 % in Brazil. These values were generated from a restricted database, in addition to the use of subjective methodologies, which makes it difficult to compare results, quantify the volume, and identify the causes more precisely (Lana, 2016). The inherent characteristics of fruits and vegetables, especially their high perishability, contribute significantly to this waste. The problem is divided into 10 % FLW in the field, 30 % in the processes related to storage and distribution, 50 % in transportation and 10 % in households (FAO, 2015), with half of this food consisting of fruit, vegetables and greens (Brazil, 2022). In addition to social and economic impacts resulting from food loss and waste, not to mention the inadequate use of soil, water, energy, pesticides and fertilizers, the emission of greenhouse gases has become increasingly alarming (FAO, 2015). Around 8–10 % of GHG emissions arise from food losses and waste (UNEP, 2024).

The annual emission of carbon dioxide (CO<sub>2</sub>eq) on the planet regarding all sectors of the economy was 34,739.12 MtCO<sub>2</sub>eq in 2022. Brazil emitted 798.61 MtCO<sub>2</sub>eq in the same year (WRI, 2023). In 2019, per capita emissions of equivalent carbon dioxide (tCO<sub>2</sub>eq) in the country were 6.88 tCO<sub>2</sub>, placing Brazil in 9th place in the rankings of the largest emitters on the planet (WRI, 2023). Without considering emissions related to deforestation, agriculture in Brazil is responsible for

9.8 % of emissions. Specifically, livestock farming accounts for 63 % of these emissions. Specifically with regard to plantations, the impacts of gas emissions are smaller than those of livestock farming, but they exist, and are mainly related to the intensive use of fertilizers (Vargas et al., 2023).

The 2030 Agenda established the 17 SDGs launched by the UN in 2015 as a challenging action plan where all interested parties must commit to sustainable development. In this sense, there is an urgent need for greater commitment from interrelated parties to reformulate their actions more sustainably with greater efficiency from food production to consumption.

Despite efforts made in food waste management, such as reuse, recycling and recovery, Papargyropoulou et al. (2014) point out prevention as being the most recommended action to reduce food waste, both concerning a better use of natural resources and their appropriate purpose, i.e. human consumption. Disposal or elimination of waste is considered the least appropriate option (Ciccullo et al., 2021; Brancoli et al., 2020; Natural Resources Defense Council – NRDC, 2020; WRAP, 2018; Hermsdorf et al., 2017; Papargyropoulou et al., 2014), in addition to the fact that environmental impacts are greater at the end of the supply chain (Schinkel, 2019; Brancoli et al., 2017).

Biermann et al. (2017) and Mena et al. (2011) emphasize a lack of accountability and the absence of clear, efficient processes and goals among the organizations involved in the food supply chain, which should be focused on formulating more sustainable solutions. Furthermore, the availability of direct primary data on food waste remains limited, and the quantification methods employed are highly inconsistent, as highlighted by the Food Waste Index Report from the United Nations Environment Program (UNEP) (UNEP, 2021).

### 2.2 Life cycle assessment

Life Cycle Assessment is a standardized methodology used to evaluate the environmental impacts of a product or service throughout its life, according to NBR ISO 14040 (ABNT, 2009a). This tool considers all phases of the life cycle, from the extraction of raw materials, energy, transport, production and use to the final disposal of a product or service. The Brazilian Association of Technical Standards establishes the principles and structure to LCA based on standards standards NBR ISO 14040 (ABNT, 2009a) and NBR ISO 14044 (ABNT, 2009b).

Human actions have impacted the rates of carbon emissions into the atmosphere, and the concentrations of greenhouse gases (GHG) are rapidly increasing thereof, which has brought serious risks to humanity (Hairiah et al., 2010). Given this scenario, the relevance of LCA to estimate the potential impacts of carbon emissions caused by food waste in supermarket retailing is justified, as it allows measuring equivalent CO<sub>2</sub> emissions. However, it should be noted that it cannot be considered an accurate prediction (Brancoli et al., 2020).

Thus, this work aims to highlight the environmental impacts caused by foods unsold at retail using knowledge about different levels of emissions of the foods analyzed through the LCA.

## 3 Materials and methods

The quantification of fruit and vegetable (FV) waste was carried out in three supermarkets in a city in the State of Sao Paulo, Brazil, belonging to different retail chains. Retail establishments were selected through convenience sampling, based on their interest and willingness to participate in the research. The names of the companies were concealed to preserve anonymity in accordance with a signed confidentiality agreement.

It is important to note that the data collection period was during the spring and summer, and it was not possible to collect in all seasons of the year, which could vary in terms of the type of fruit or vegetable since some products are seasonal. Additionally, the collection was concentrated over one week in each store. Despite these limitations, the amount

of food waste, considering the most wasted products, corresponded to the perspective of store employees and managers who had already reported their perceptions.

A waste composition analysis was carried out aiming to identify, sort, weigh and categorize unsold food from these companies during a period of seven calendar days in each store. All discarded FV from the stores were analyzed, including those removed from sales areas for failing to meet aesthetic standards, those deteriorated during arrangement, transportation, or storage, and waste generated in handling areas used for repurposing into snacks, pies, cakes, or fruit salads. According to the 2009 UK WRAP (Waste and Resources Action Programme) Report, developed by [Quested and Johnson \(2009\)](#), food waste can be defined in three ways: *avoidable waste*, possibly *avoidable waste* and *unavoidable waste*: *Avoidable waste* is that which is characterized by food discarded due to actions that could have been taken before disposal, for example, expired products. *Possibly avoidable waste* refers to food that is consumed by some people and not by others (such as apple peels, for example). *Unavoidable waste* includes the remains of the preparation of food or drinks that are inedible in normal situations (pits, seeds and peels) ([Quested, Johnson, 2009](#); [UNEP, 2021](#); [Liu et al., 2024](#)). In this study, there was no distinction between avoidable, possibly avoidable and unavoidable parts of fruits and vegetables. Only a small fraction of stalks, roots and stems were considered as unavoidable foods.

After the waste composition analysis, an LCA was performed to estimate the amounts of CO<sub>2</sub>eq emitted by the measured waste, given the relevance of not only measuring FV waste in terms of weight, but also by estimating the impacts exerted on account of environmental problems caused by waste, as pointed out by [Scholz et al. \(2015\)](#) and [Dreyer et al. \(2019\)](#).

The LCA process consists of four main phases: (i) goal and scope definition, where the purpose of the study and system boundaries are established; (ii) life cycle inventory, which involves the collection and quantification of input and output data related to the product system; (iii) life cycle impact assessment, where potential environmental impacts are calculated based on the inventory data; and (iv) interpretation, which evaluates results in relation to the defined goals, enabling the identification of key contributors and improvement opportunities (NBR ISO 14040). This structured approach ensures a comprehensive understanding of the environmental burdens associated with food waste in the retail sector.

[Table 1](#) presents a summary of the methodological procedures used in the study.

For better visualization, the methods are presented individually as follows.

3.1 Waste composition analysis

After registering the unsold FV in their internal systems, the companies made all these foods available for analysis of their composition. Food was stored in plastic boxes whence it was removed, sorted, categorized and recorded using a control spreadsheet. Afterwards, products were placed in other plastic boxes located in disposal areas identified by companies. A digital scale was used with accuracy of two grams and

maximum capacity of 15 kg.

In this study, all food identified during the waste composition analysis was regarded as waste, including avoidable food waste consisting of edible food discarded before being consumed, and unavoidable waste which are inedible parts under normal circumstances (such as seeds and peels) ([Liu et al., 2024](#); [UNEP, 2023](#)).

For the purposes of performing the LCA, foods under study were divided into five categories: tubers, bulbs and roots; processed products; unavoidable; fruits; and leaves, flowers and stems in accordance with a waste categorization method developed by [Brancoli et al. \(2022\)](#). Studied companies have different sizes, reflecting important characteristics for the handling, conservation and sale of products.

Store 1 has a FV sales area of 238.18 m<sup>2</sup>, Store 2 has 26.79 m<sup>2</sup> and Store 86.44 m<sup>2</sup>. In Store 1, six operational employees and one FV manager were responsible for the sector. In Store 2, there was only one operational employee and one FV manager. In Store 3, there were three operational employees and two managers in the sector.

Some of the studied products had been processed (such as trays of chopped vegetables and ready-made salads) and arrived at the waste composition analysis location mixed with whole products and without packaging. The same occurred with waste from rotisseries and bakeries. However, a composition analysis of these wasted products was carried out in detail to correctly identify the foods and subsequently categorize and quantify them.

After the waste composition analysis, the food followed their organizations' regular routine, i.e. it was initially stored in warehouses before its final disposal. Therefore, none of the food identified in the analysis was redirected to its primary purpose of human consumption.

3.2 Life cycle assessment

The objective of this research is to assess the environmental impact, particularly the carbon footprint, of food waste generated in three supermarkets located in the state of Sao Paulo, Brazil. The goal is to estimate the greenhouse gases emissions (expressed as kg of CO<sub>2</sub> equivalent) associated with FV waste, enabling the identification of key contributors, or "hotspots," that have the most significant impact on the climate. These findings are intended to guide future strategies aimed at reducing food waste and mitigating its environmental effects through targeted interventions such as waste prevention and valorisation.

The functional unit for this study is defined as the total amount of food waste generated per supermarket over a specified one-week period. The system boundary for the study is cradle to grave, meaning it includes all stages from food production, distribution, and retail to the final disposal of waste.

Agricultural production processes, such as land use, seedling growth, application of fertilizers and pesticides, machinery operations, energy use, and transportation, are included in the analysis. For processed food items, additional stages such as packaging and food preparation are also considered. The unavoidable food waste fraction (e.g., inedible parts like peels and seeds) is excluded from the carbon impact of food production since these parts are byproducts of the edible portion, which has already been consumed, and only the environmental impact of the end-of-life management for these unavoidable fractions is considered. The modeling assumed herein is based on that used by [Liu et al. \(2024\)](#) in their study on presenting the carbon footprint of food waste from street markets, also carried out in Brazil. Variations in fertilizer use, irrigation, electricity use, seedling production, machinery and other processes were inventoried to represent local conditions. The remaining data for agricultural production, including transport, was retrieved from the ecoinvent database ([Wernet et al., 2016](#)).

Transportation distances for food products were not directly modeled but are implicitly included within the market activities used from the ecoinvent database ([Wernet et al., 2016](#)). In ecoinvent, market activities represent the average consumption mix of a product and include transportation steps from the point of production to the point of

**Table 1**  
Methodological procedures.

| Descriptive research    |                                          |
|-------------------------|------------------------------------------|
| Object of analysis      | Three stores in the supermarket sector   |
| Analysis location       | A city of countryside of Sao Paulo State |
| Study period            | Store 1: from 19 to 26/11/2022           |
|                         | Store 2: from 01/26–02/01/2023           |
|                         | Store 3: from 17 to 22/03/2023           |
| Data collection methods | Survey and document analysis             |
| Data analysis procedure | Descriptive statistics                   |
|                         | LCA                                      |

Source: The authors.

consumption. These datasets account for transport distances and modes (e.g., road, rail, sea) based on regional trade flows and product-specific logistics. As such, the transportation of the food products are embedded in the selected market datasets.

The impact assessment is based on the global warming potential, using the ReCiPe 2016 midpoint method (Huijbregts et al., 2016), as implemented in the SimaPro software.

Results of this study are presented below.

## 4 Results

### 4.1 Waste composition analysis and life cycle assessment

The waste composition analysis quantified 2160.3 kg of products unsold by retailers. 783.2 kg of discarded food were identified in Store 1, 234.9 kg in Store 2 and 1142.2 kg in Store 3. These foods were categorized, sorted and weighed.

The ten main foods unsold from 64 items in Store 1 (bananas, cassavas, apples, tomatoes, potatoes, oranges, melons, cabbages, carrots and cucumbers) totaled 426.6 kg of food corresponding to 54.4 % of the supermarket's total amount of waste (Table 2).

In Store 2, 39 different items were recorded, and the top ten wasted products (bananas, tomatoes, oranges, mangos, carrots, cabbages, avocados, papayas, eggplants and potatoes) amount to 183.5 kg corresponding to around 78.1 % of the total food wasted in this store (Table 3).

In Store 3, gravimetric data revealed 55 wasted items, especially the principal foods (bananas, cabbages, tomatoes, potatoes, oranges, pumpkins, apples, mangoes, lemons and tangerines) which together accounted for 847.1 kg and amount to 74.1 % of total food waste (Table 4).

The high wastage of cabbage found in the waste composition analysis, mainly at Store 3, can be explained by the removal of outer leaves of these foods upon receipt, as well as during their display in stalls. This happens to maintain a good aspect of products given that, according to Brancoli et al. (2022), the conditions in which this type of product is found during sales (unit, temperature) and the handling of products by consumers cause damage to outer leaves which are removed if they are not in optimal conditions. Authors such as Liu et al. (2024), Mijares et al. (2021), Ortiz-Gonzalo et al. (2021), Lana and Moita (2019), Nogueira et al. (2020) and Eriksson et al. (2012) also report large amounts of leafy waste.

In the case of tomatoes discarded in large quantities, it was found that some of the products delivered to the store were in closed packages. The use of packaging can be a disadvantage in relation to FV waste, as

pointed out by Mattsson and Williams (2022). This is because, for instance, a spoiled tomato inside the package implies the disposal of the entire packaging, without any type of reuse of perfect products, due to the impossibility of replacing the product and sealing the packaging once more. The same problem extends to other packaged products, such as apples, pears, oranges and carrots, for example.

It was observed that almost half of the food surplus found in the study (1 ton) was composed of the following foods: banana (383.7 kg), tomato (189.7 kg), cabbage (184.3 kg), potato (123.9 kg) and orange (119.8 kg). These same foods are the most largely produced crops in the State of São Paulo, occupying its largest areas, as aforementioned (IBGE, 2022). It is therefore observed that some of the most produced foods are also the ones which are most wasted.

Bananas stood out from other products, accounting for approximately 18 % of total food waste and were the most wasted food in the three companies. Several studies indicate that this product is the most wasted food around the planet (Liu et al., 2024; Santos et al., 2020; Mattsson et al., 2018; Brancoli et al., 2017; Eriksson, 2015; Scholz et al., 2015) and are the most consumed fruit in Brazil (CES/IBGE, 2020). According to Riley (2014), one of the main causes of banana waste is the consumer's lack of knowledge about the shape or imperfect ripening of the product.

The amounts of tomatoes and cabbages wasted was similar, in terms of mass, in the stores studied. Other studies carried out in different countries highlight the foods researched herein as being the most wasted as well, such as in Cakar (2022) (tomatoes, potatoes and bananas); Jun et al. (2022) (bananas and leafy vegetables), Mattsson and Williams (2022) (potatoes, bananas, tomatoes and oranges); Bilska et al. (2018) (tomatoes, cabbages, potatoes and bananas); Mattsson et al. (2018), Eriksson (2015) and Scholz et al. (2015) (bananas and tomatoes). This evidences that the problem of wasting these products occurs on a large scale and requires greater attention from retailers regarding the operational management of their stores.

All FV waste quantified in the waste composition analysis, totaling 2160.3 kg, was modeled in the LCA based on the available gravimetric data. 878.1 kg of CO<sub>2</sub>eq were estimated for the waste of 783.2 kg of FV in Store 1; 252.7 kg of CO<sub>2</sub>eq for 234.9 kg of food wasted in Store 2; and 1235.5 kg of CO<sub>2</sub>eq emissions were calculated for 1142.2 kg of food wasted in Store 3.

Fig. 1 reveals the amounts of wasted food and those of equivalent carbon dioxide (CO<sub>2</sub>eq) emitted by each category of food analyzed in each of the three stores under study, both during production and at their End of Life - EoL.

A contribution analysis was performed to identify which categories and products were responsible for the highest share of emissions.

**Table 2**  
Food wasted in Store 1.

| Total waste Store 1 |      |               |      |                  |     |                    |     |
|---------------------|------|---------------|------|------------------|-----|--------------------|-----|
| Food                | Kg   | Food          | Kg   | Food             | Kg  | Food               | Kg  |
| Banana              | 77,1 | Plum          | 16,9 | Spinach          | 5,6 | Maroon cucumber    | 1,1 |
| Cassava             | 51,0 | Pumpkin       | 16,7 | Avocado          | 3,6 | Chicory            | 1,1 |
| Apple               | 48,3 | Mango         | 15,1 | Strawberry       | 3,4 | Okra               | 0,9 |
| Tomato              | 44,9 | Eggplant      | 14,0 | Onion peels      | 3,0 | Fresh fennel       | 0,8 |
| Potato              | 40,9 | Lemon         | 13,7 | Cauliflower      | 2,9 | Guava              | 0,7 |
| Orange              | 37,4 | Peach         | 12,7 | Watercress       | 2,8 | Celery             | 0,7 |
| Melon               | 35,1 | Grape         | 11,7 | Passion fruit    | 2,8 | Persian lime       | 0,5 |
| Cabbage             | 33,8 | Chayote       | 10,3 | Arracacha        | 2,7 | Leek               | 0,5 |
| Carrot              | 31,0 | Watermelon    | 9,9  | Ginger           | 2,3 | Mint               | 0,5 |
| Cucumber            | 27,1 | Broccoli      | 7,8  | Arugula          | 2,3 | Radish             | 0,2 |
| Green pepper        | 27,1 | Corn          | 7,4  | Parsley          | 2,0 | Blackberry         | 0,2 |
| Papaya              | 26,8 | Collard green | 6,5  | Coconut          | 2,0 | Basil              | 0,2 |
| Pear                | 23,7 | Escarole      | 6,1  | Chives           | 2,0 | Parsley and chives | 0,2 |
| Lettuce             | 22,0 | Pepper        | 6,0  | Scarlet eggplant | 2,0 | Kiwi               | 0,2 |
| Beet                | 21,8 | Purple yam    | 5,9  | Sweet potato     | 1,5 | Yam                | 0,1 |
| Tangerine           | 18,3 | String bean   | 5,8  | Chinese cabbage  | 1,5 | Laurel (branches)  | 0,1 |

Source: The authors based on gravimetric data.

**Table 3**

Food wasted in Store 2.

| Total waste at Store 2 |      |               |     |                     |     |                 |     |
|------------------------|------|---------------|-----|---------------------|-----|-----------------|-----|
| Food                   | Kg   | Food          | Kg  | Food                | Kg  | Food            | Kg  |
| Banana                 | 46,7 | Cucumber      | 6,9 | Lemon               | 2,2 | Chicory         | 0,5 |
| Tomato                 | 37,1 | Beet          | 5,0 | Ginger              | 1,5 | Arracacha       | 0,5 |
| Orange                 | 26,1 | Pepper        | 4,7 | Scarlet eggplant    | 0,9 | Chinese cabbage | 0,5 |
| Mango                  | 15,1 | Watermelon    | 4,1 | Dandelion green     | 0,9 | Grape           | 0,5 |
| Carrot                 | 14,9 | Peach         | 4,0 | Plum                | 0,8 | Leek            | 0,4 |
| Cabbage                | 12,4 | Apple         | 3,5 | Lettuce             | 0,8 | Green pepper    | 0,2 |
| Avocado                | 8,4  | Pear          | 3,1 | Arugula             | 0,8 | Chives          | 0,1 |
| Papaya                 | 8,0  | Pumpkin       | 2,6 | Okra                | 0,6 | Onion           | 0,1 |
| Eggplant               | 7,7  | Passion fruit | 2,6 | Broccoli            | 0,6 | Parsley         | 0,1 |
| Potato                 | 7,1  | Chayote       | 2,4 | Purple sweet potato | 0,5 |                 |     |

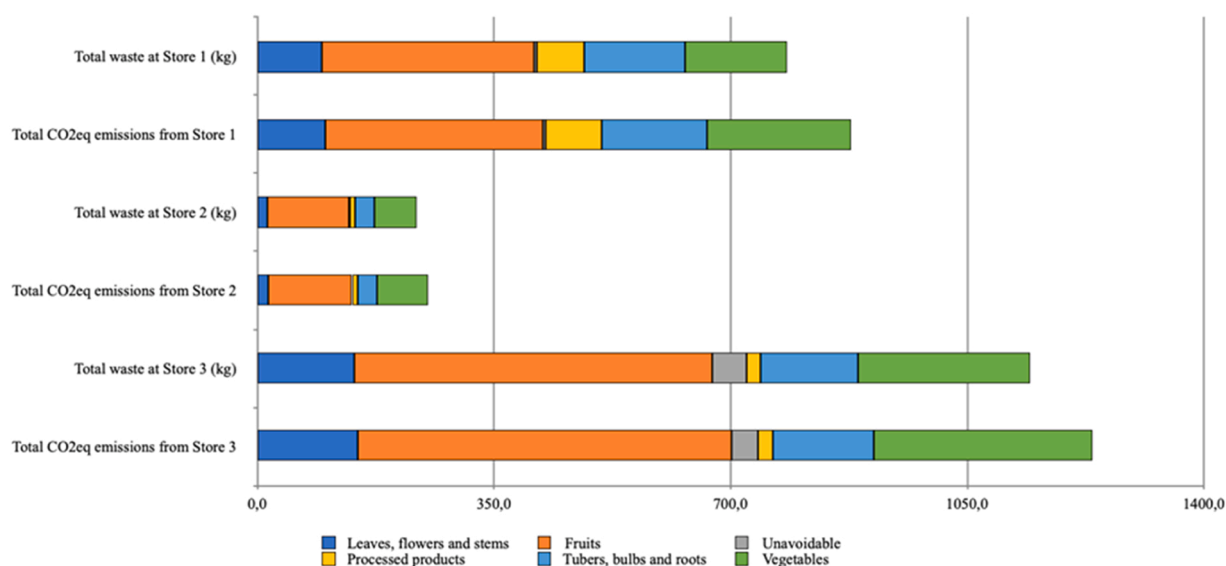
Source: The authors based on gravimetric data.

**Table 4**

Food wasted in Store 3.

| Total waste at Store 3 |       |                  |      |                     |     |         |     |
|------------------------|-------|------------------|------|---------------------|-----|---------|-----|
| Food                   | Kg    | Food             | Kg   | Food                | Kg  | Food    | Kg  |
| Banana                 | 259,9 | Carrot           | 22,0 | Melon               | 3,6 | Yam     | 1,5 |
| Cabbage                | 138,1 | Pepper           | 18,4 | Lettuce             | 3,2 | Chicory | 1,3 |
| Tomato                 | 107,7 | Passion fruit    | 18,0 | String bean         | 3,2 | Ginger  | 1,3 |
| Potato                 | 75,9  | Papaya           | 14,3 | Pepper              | 3,2 | Chives  | 1,2 |
| Orange                 | 56,3  | Watermelon       | 12,3 | Garlic              | 3,0 | Grape   | 0,9 |
| Pumpkin                | 55,3  | Corn             | 11,8 | Chinese cabbage     | 2,8 | Guava   | 0,9 |
| Apple                  | 42,4  | Chayote          | 9,7  | Kiwi                | 2,3 | Okra    | 0,8 |
| Mango                  | 38,3  | Coconut          | 6,7  | Avocado             | 2,3 | Arugula | 0,5 |
| Lemon                  | 36,7  | Beet             | 6,6  | Star fruit          | 2,2 | Peach   | 0,5 |
| Tangerine              | 36,5  | Broccoli         | 6,3  | Purple sweet potato | 2,2 | Parsley | 0,4 |
| Onion                  | 27,3  | Persimmon        | 5,8  | Nectarine           | 2,0 | Fig     | 0,3 |
| Cucumber               | 27,1  | Pineapple        | 5,0  | Dandelion green     | 1,7 | Pear    | 0,2 |
| Arracacha              | 25,6  | Scarlet eggplant | 4,7  | Pitaya              | 1,7 | Plum    | 0,2 |
| Eggplant               | 24,9  | Strawberry       | 3,6  | Radish              | 1,6 |         |     |

Source: The authors based on gravimetric data.

**Fig. 1.** Amounts of wasted food and equivalent carbon dioxide emissions (CO<sub>2</sub>eq - EoL), estimated for each food category analyzed at the three stores. Source: The authors according to the LCA results.

According to available data, the LCA resulted in 2366.3 kg of CO<sub>2</sub>eq emitted by the waste analyzed. Total emissions per category were estimated at around 1000.1 kg CO<sub>2</sub>eq for fruits, corresponding to 42.3 % of overall emissions; 612.3 kg CO<sub>2</sub>eq for vegetables, constituting to 25.9 % of total emissions; 334.1 kg of CO<sub>2</sub>eq for tubers, bulbs and roots, representing 14.1 % of total emissions; 263.5 kg of CO<sub>2</sub>eq for leaves, flowers and stems, accounting to 11.1 % of total emissions; 113,9 kg of

CO<sub>2</sub>eq for processed products, which amounts to 4.8 % of total emissions; and 42.4 kg of CO<sub>2</sub>eq for unavoidable food, making up to 1.8 % of total footprint (Fig. 2).

The end-of-life stage is a major contributor to the carbon footprint across all product categories, accounting for approximately 35–70 % of total life cycle emissions depending on the product group. This is primarily due to the fact that, in Brazil, landfilling is the predominant waste

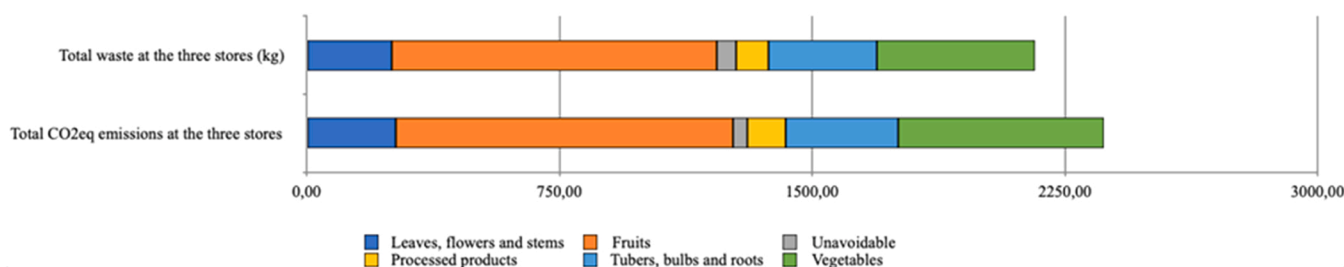


Fig. 2. Amounts of wasted food and equivalent carbon dioxide emissions (CO<sub>2</sub>eq - EoL) estimated at the three stores. Source: The authors according to the LCA results.

management practice (Lino et al., 2023).

The most wasted foods in each category analyzed, which contribute the most to climate change, are tomatoes, peppers and cucumbers (vegetables); bananas, oranges and apples (fruits); cabbage and broccoli (leaves, flowers and stems); and potatoes, cassava and carrots (tubers, bulbs and roots). Together, these products are responsible for more than 60 % of the emissions from the total waste analyzed. Brancoli et al. (2017) found that tomatoes and bananas waste were the foods responsible for the highest CO<sub>2</sub> emissions in the fruit and vegetable category. In their study, bananas and tomatoes, together with strawberries, corresponded to 70 % of emissions from the fruit and vegetable waste fraction in a Swedish supermarket (Brancoli et al., 2017).

Considering the total emissions, in relation to the quantities of wasted food, there is an average estimate of 1.3 CO<sub>2</sub>eq/kg of vegetables; 1.2 CO<sub>2</sub>eq/kg of processed products; 1 CO<sub>2</sub>eq/kg of leaves, flowers and stems, fruits, tubers, bulbs and roots; and 0.8 CO<sub>2</sub>eq/kg of unavoidable. Therefore, vegetables constitute the category responsible for the highest emission of CO<sub>2</sub>eq/kg, even surpassing the category of processed foods. It must be considered that the products analyzed in this study are fresh products or products with minimal processing (cut, chopped, sliced or grated), not including products with a high degree of industrialization, such as canned vegetables.

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Our results, which show that FV constitute the largest portions of waste in this study (approximately 66 %), align with findings by Albalate-Ramírez et al. (2024), who reported a similar share (51.3 %) in the Mexican food sector, and, consequently, account for greater CO<sub>2</sub> emissions, attention to these critical categories is highlighted as a priority, since they reflect significant economic and environmental loss (Albalate-Ramírez et al., 2024).

Fruit was the avoidable food which was wasted the most in all companies under study, representing 44.6 % of the total waste identified in the research, which is also accountable for 42.3 % of estimated emissions from all waste identified. Although emissions in the production of each food vary, their end-of-life emissions were modeled using the same approach for landfilling organic waste (0.76 kg CO<sub>2</sub>eq/kg), considering disposal in landfills as the final disposal of food, which is rather common in developing countries (Lino et al., 2023). Such disposal of waste increases the carbon footprint (Liu et al., 2024), emphasizing the need to improve operational food management and achieve its main purpose, i.e. human consumption, or at least adequate reuse of waste for other purposes, such as animal feed, composting and energy generation (Papargyropoulou et al., 2014).

If the amount of food wasted and estimated emissions were extrapolated for one year, assuming that the same amounts and emissions were generated during each week of the year, the three companies altogether would be responsible for a total of 113 tons of waste and emit almost 123.5 tCO<sub>2</sub>eq. With the exception of the unavoidable parts found in the

waste composition analysis, around 110 tons of food could be salvaged, reused or redistributed, which would avoid the emission of 109.2 tCO<sub>2</sub>eq.

#### 4.2 Managerial and infrastructural implications of fv waste

Given the perishability of analyzed products, it is worth mentioning that the climate of the region in which the present study was carried out greatly affects FV waste (Cicatiello et al., 2020; Parfitt et al., 2010; Ludwig-Ohm et al., 2019; Hudson and Messa, 2014; Mena et al., 2011). The temperature in the region where companies were located is rather high, requiring refrigeration for most studied products. During the week analysis in Store 1, the average temperature in the region was 25°C. However, the temperature variation was quite drastic, a common characteristic of the municipality, with temperatures ranging from 19°C to 31°C, according to the Meteored (2024) website. During the measurement period in Store 2, the average temperature was 26°C, ranging from 21°C to 31°C. Meanwhile, in Store 3, the average temperature was 25°C, ranging from 19°C to 31°C.

Despite these elevated temperatures, only Store 1 had a cold room of FV storage, and the entire store was air-conditioned. Store 2 did not have a cold room, nor refrigeration in the sales area and the remainder of the store. Stocked products were stored at the back of the company in a partially covered space. Thus, products were exposed to sunlight, heat and humidity. Store 3 had a small cold room, which not only failed to meet the storage needs for all refrigerated products but was also used to display and sell other types of merchandise, such as dairy products. The store was air-conditioned during the day, but the units were turned off at night, exposing products to sudden temperature fluctuations. FV were stored alongside other grocery items in a non-refrigerated warehouse, further compromising their preservation.

The problem of lack of refrigeration is reported in literature as one of the main reasons for the loss of freshness of food, as pointed out by Liu et al. (2024), Mattsson and Williams (2022), Cicatiello et al. (2020); Parfitt et al. (2010) and Nunes et al. (2009). Ribeiro et al. (2019) highlight that refrigerated storage space is insufficient for the quantity of perishable and fresh products offered by retail stores and, according to Porat et al. (2018), retail lacks adequate refrigeration infrastructure, in addition to the fact that changes in temperature result in a reduction of the shelf life of food. Among the 3 stores studied in this research, this was one of the causes of waste.

A recent study on the leafy vegetable chain, carried out in Brazil, highlighted, among other factors, the need for infrastructure improvements in the chain of these foods to mitigate waste, given the impact of climate conditions in Latin American countries (Realpe et al., 2024). In this same context, Santos et al. (2020) investigated storage in sales areas and found that inadequate temperatures and conditions, and hot weather influence the conservation of fruit and vegetables.

The food retail sector faces the challenge of inventory management, especially of perishable products. This is because food that is not sold at the end of the day needs to be discarded, while if there is a shortage of these items, companies lose the opportunity to sell and, therefore,

determining the adequate stock suffers from fluctuating demand (Sakoda et al., 2019). This can be seen in the three stores studied: inadequate planning of fruit and vegetable purchases, not considering historical consumption data, sales history and waste when planning purchases.

In Brazil, leafy vegetables are commonly sold under take-back agreements, (Realpe et al., 2024), where suppliers are only paid for products sold, and unsold items are collected by the suppliers, who assume the financial cost and handle final disposal. This arrangement exempts supermarkets from waste-related costs and responsibilities for maintaining product quality on shelves. Two of the stores studied (Stores 2 and 3) operate under these agreements. This fact could help explain the differences in the proportion of leafy vegetables in total waste compared to the findings of Brancoli et al. (2022), who observed much higher quantities of leafy vegetables, such as lettuce, in street markets where take-back agreements were not present.

Another problem found in the three stores was the piling of food in storage areas, especially in smaller companies due to lack of space. Although stores receive products packed in plastic boxes suitable for stacking ("type K" boxes), other types of unsuitable boxes were found (made of pine wood or cardboard). Some products, such as oranges, lemons and potatoes, were delivered in bags and stored under the same conditions, which clearly compromised food preservation.

Stacking was also recorded in the display of food on stalls and shelves (sales area). In this regard, only Store 1 used a false bottom of stalls as a strategy to avoid too much stacking of products and consequent deterioration. The other two stores had the practice of stacking several layers of products piled up in deep stalls.

Regarding reuse strategies, only Stores 1 and 3 used visually imperfect or damaged products, which were cut and repackaged, and sent back to the sales area, when there was availability of labor. Store 2 was barred by the health sanitary surveillance from handling food on account of not being air-conditioned. Price reduction strategies aimed at mitigating waste, which is a common practice in supermarket retail (Filimonau and Gherbin, 2017; Lebersorger and Schneider, 2014), were not observed in stores. Products were regularly on sale, especially on FV promotion days (Tuesdays and Wednesdays), without differentiating products in more advanced stages of ripeness or having other damages.

The relation between the amount of food waste and the size of stores was not verified, as in the research of Teller et al. (2018) and Lebersorger and Schneider (2014). Store 3 was the one with the most waste, although it is ranked second in size, followed by Store 1, the largest company, and Store 3, the smallest. Store 1 demonstrated better performance only in relation to waste considered by sales area ( $m^2$ ), different from the result in relation to the number of employees and the variety of foods sold. Store 3, the smallest of the companies analyzed, even without refrigeration systems in the store, had better performance in terms of waste relative to the number of employees and the variety of foods sold. Furthermore, only Store 1 reported offering adequate training to employees for handling FV. In the other companies, learning occurs in practice, on a day-to-day basis, through observation. This demonstrates the difficulty of comparing companies, each with its own peculiar characteristics.

Although the objective of this research was not to judge the actions carried out by the companies studied, it is possible to observe the low interest of managers in prioritizing actions that avoid or reduce waste of FV since the sector considers waste of FV a common consequence in supermarkets, due to the type of product sold.

The results of this work can be useful for supermarket retailers to consider greater commitment in developing initiatives able to contribute to reducing environmental impacts caused by food waste, considering foods wasted the most and exerting the greatest environmental impact. Strategies such as environmental labeling, i.e. including the carbon footprint corresponding to all phases of product life cycle on packaging, for instance, can have a positive impact on informing and raising awareness among consumers, especially in relation to preventing and

reducing food waste, as seen in countries such as the United Kingdom (Zhao et al., 2015). These initiatives can also be attractive for consumers, i.e. from choosing which products to buy to minimize impacts in terms of carbon footprint to better planning for purchasing and consuming food aiming to avoid food waste.

In addition to raising environmental awareness among their customers, retailers can implement other actions, such as adequate purchasing and inventory planning for companies (Audet and Brisebois, 2019), sales optimization and management strategies, including greater precision in orders (demand forecast) (Alamar et al., 2017; Horós and Ruppenthal, 2021), making specifications for non-conformity of products more flexible, such as the size, weight or color of food (Magalhães et al., 2018) and the culture of abundance in establishments (Henz and Porpino, 2017; Moraes et al., 2022), interaction with other agents, mainly in joint planning processes involving suppliers and/or distributors (Mena et al., 2011; Ramírez et al., 2021), coordination and information sharing (Magalhães et al., 2018) and data on sales planning for the consumer (Porpino et al., 2016; Stangherlin and Barcellos, 2018). These are strategic in preventing and reducing FV waste and corroborate the best recommendation of the food waste hierarchy proposed by Parpagyropoulou et al. (2014) based on the European waste hierarchy. Furthermore, other actions are also important to avoid food disposal, such as donations (Mena et al., 2011), reuse, recycling and recovery (Parpagyropoulou et al., 2014).

## 5 Final considerations

The LCA results revealed the carbon footprint associated with FV waste and identified the products whose waste contributed most to overall emissions. Regarding retail operations related to FV, several inadequate situations were observed regarding arrival, handling, storage and disposal of FV, which might have contributed to food waste, in addition to difficulties in demand forecasting. Companies, especially Store 1, have specific food quality criteria upon delivery, which can significantly reduce the level of pre-storage waste, as reported in studies by Ribeiro et al. (2019).

This study presents direct data regarding the reality of food waste in Brazilian supermarket retailing and estimated amounts of  $CO_2eq$ . Understanding the quantities of food waste and the resulting carbon footprint can serve to identify priority targets and establish efficient measures to combat the problem. A collaboration of all links in the food supply chain in combating food loss and waste (FLW) is important, with emphasis on supermarket retailing as a link exerting a strong influence, both upstream and downstream of the food chain, which can contribute to significant changes in this regard.

Nonetheless, this analysis is limited to the quantification of fresh or minimally processed fruits and vegetables (cut, chopped, sliced or grated), either packaged or not, during a period of one week at each of the three stores. Due to the data collection period of one week, it was not possible to observe potential seasonal variations in climate and demand, which may influence the amount and types of food wasted (Porat et al., 2018). The seasonality of some foods might also have influenced the identification of the largest wasted quantities of certain products wasted, as pointed out in the studies by Ribeiro et al. (2019). Furthermore, packaging materials, such as cardboard and various types of plastic, were also not considered while calculating carbon footprint.

No pre-supermarket waste was recorded, i.e. when products were delivered to stores during the study. However, this type of waste may be significant, as it has been shown to account for considerable amounts of food waste in the Swedish retail sector (Eriksson, 2015; Brancoli et al., 2017).

## CRedit authorship contribution statement

**Pedro Brancoli:** Writing – review & editing, Software, Formal analysis, Data curation. **Andréa Rossi Scalco:** Writing – review &

editing, Supervision, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Suzana Márcia Marangoni**: Writing – original draft, Visualization, Validation, Investigation, Formal analysis, Data curation.

## Funding

This work was supported by the Sao Paulo Research Foundation (FAPESP) (grant number 2021/10489–3). This study was financed in part by the Coordenação de Aperfeiçoamento de Pessoal de Nível Superior – Brasil (CAPES) - Finance Code 001.

## Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: ANDREA ROSSI SCALCO reports financial support was provided by State of Sao Paulo Research Foundation. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

Data will be made available on request.

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